Quantum Key Distribution

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Introduction to Cryptography

Confidentiality

Protects information by transforming it into secure formats accessible only to intended recipients.

Integrity

Ensures data remains unaltered during transmission using mathematical algorithms.

Authentication

Verifies identity and provides non-repudiation for secure digital transactions.

Modern cryptography secures online communication, digital payments, cloud storage, and countless everyday applications through advanced mathematical techniques.

Why Cryptography Matters

As data travels across networks, it becomes vulnerable to hacking, theft, and manipulation. Cryptography ensures confidentiality, integrity, and authenticity by converting data into secure, unreadable formats.

It safeguards communication, online transactions, personal data, and organizational information—making it a critical component of modern cybersecurity infrastructure.

3

Core Functions

Confidentiality, integrity, authentication

Evolution of Cryptographic Methods

1 Traditional Cryptography

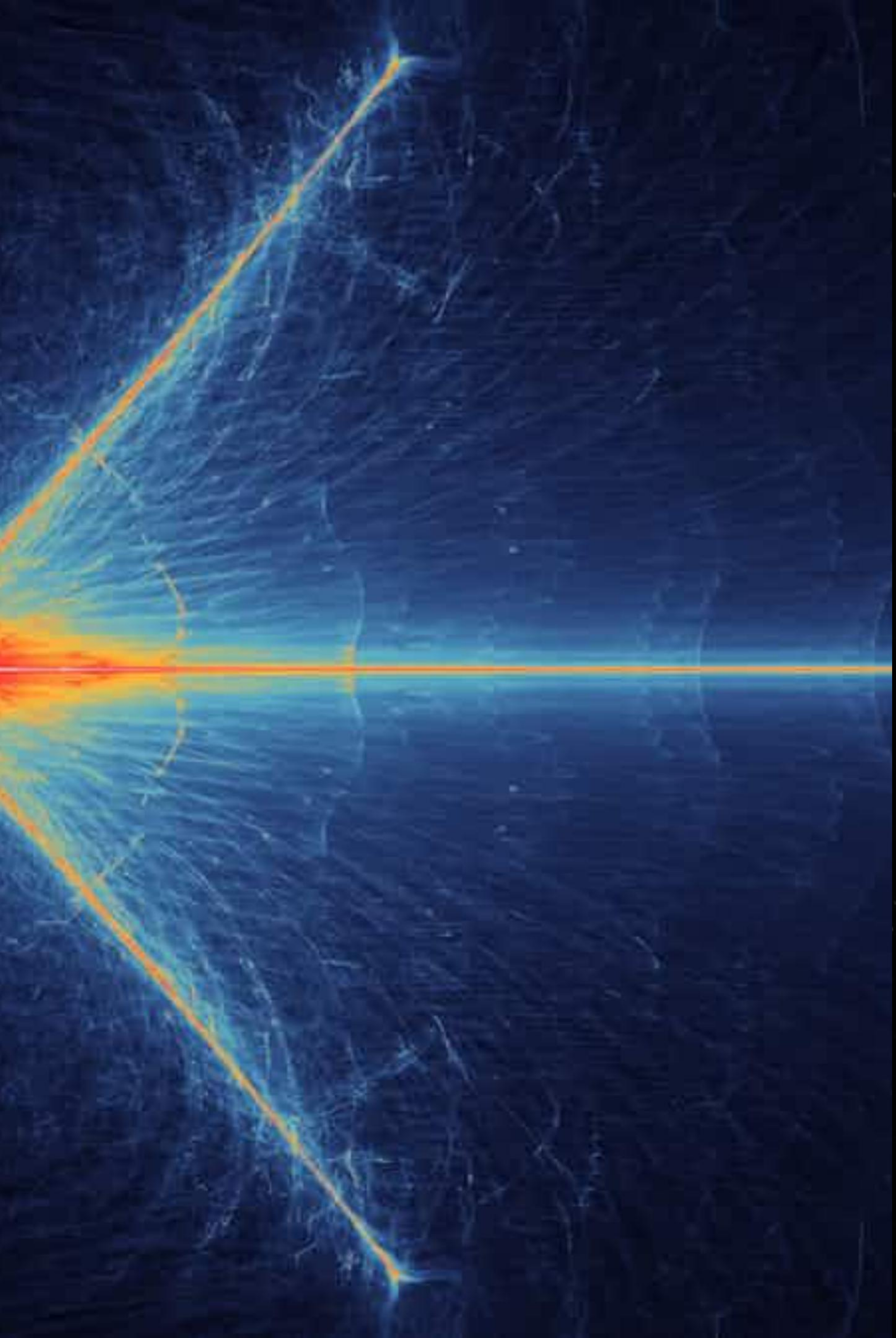
Ancient methods using substitution ciphers, transposition ciphers, and Caesar cipher with manual techniques and shared secret keys.

2 Modern Cryptography

Computer-based algorithms using complex mathematical operations for secure digital communication.

3 Quantum Cryptography

Uses quantum mechanics principles and photon particles to detect eavesdropping through state changes.



What is Quantum Key Distribution?

Quantum Key Distribution (QKD) is a secure communication technology that uses quantum mechanics principles to create and share encryption keys. Unlike traditional methods, QKD detects any eavesdropping because measuring quantum particles changes their state.

Quantum Detection

Any interception attempt immediately reveals itself through quantum state changes.

Future-Proof Security

Powerful tool for networks, banking, and government communication systems.

Quantum Mechanics Principles in QKD

Superposition

Quantum states encode information in multiple states simultaneously.

Heisenberg's Uncertainty

Measurement disturbs quantum states, revealing eavesdroppers.

Physics-Based Security

Security guaranteed by laws of physics, not computational complexity.

No-Cloning Theorem

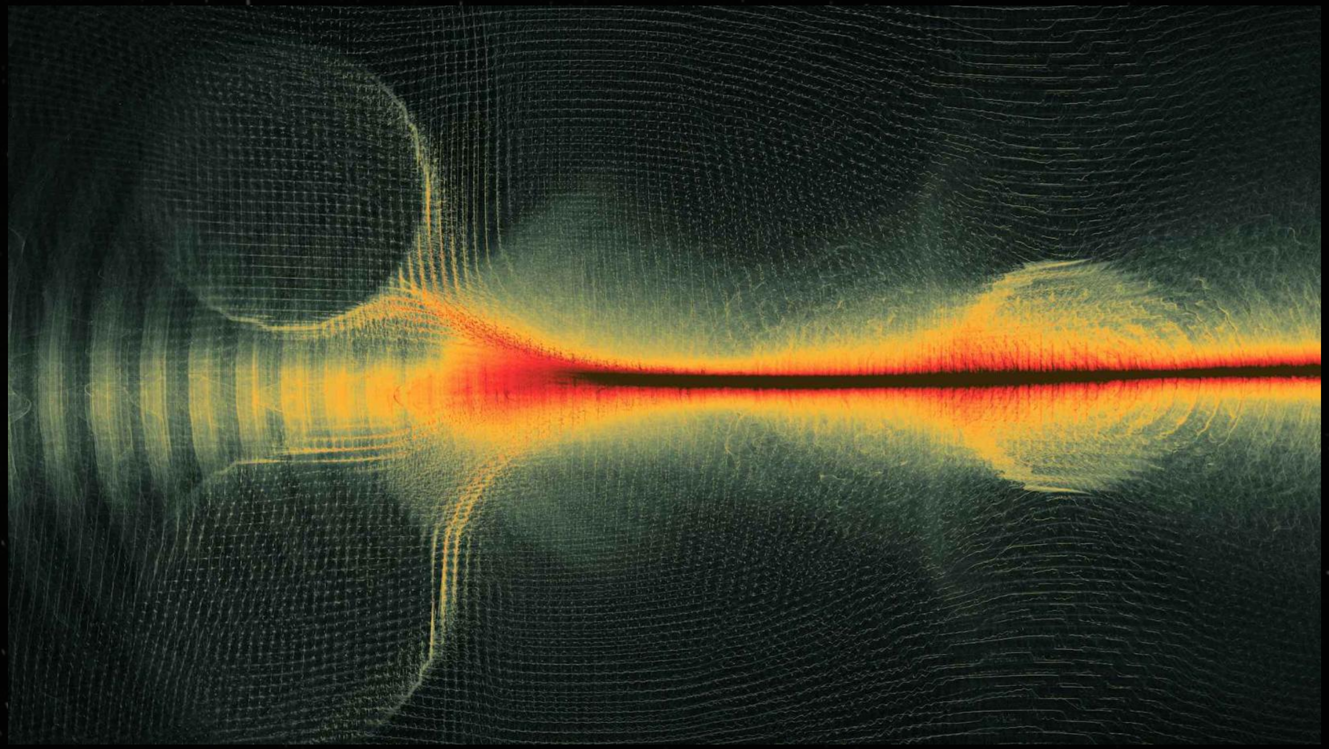
Quantum information cannot be copied, ensuring security.

Qubits and Photon Polarization

Quantum Bits

Unlike classical bits (0 or 1), qubits exist in **superposition** of both states simultaneously, enabling efficient processing of complex information.

Photon **polarization**—the direction light waves vibrate—represents qubits. Horizontal and vertical polarizations encode 0 and 1, with combinations forming superpositions.



No-Cloning Theorem: It's impossible to create an exact copy of an unknown quantum state. Unlike classical information, quantum information cannot be perfectly cloned due to quantum mechanics linearity—protecting communication security.

BB84 Protocol: Step-by-Step

The BB84 protocol, introduced by Charles Bennett and Gilles Brassard in 1984, is the first quantum key distribution method. It uses photons polarized in different bases to ensure any eavesdropping disturbs quantum states and is immediately detected.

01

Photon Transmission

Sender transmits quantum bits encoded in photon polarization states.

02

Measurement

Receiver measures photons to generate shared secret key.

03

Key Sifting

Filters out mismatched measurements from different bases.

04

Error Correction

Detects and fixes errors from noise or eavesdropping.

05

Privacy Amplification

Transforms data into shorter, highly secure output to ensure confidentiality.

Software Model Results

16

Total Photons

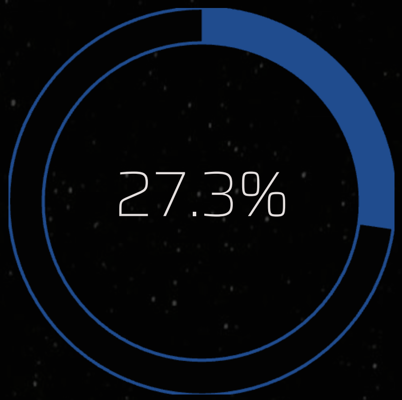
11

Bases Matched

69%

Efficiency

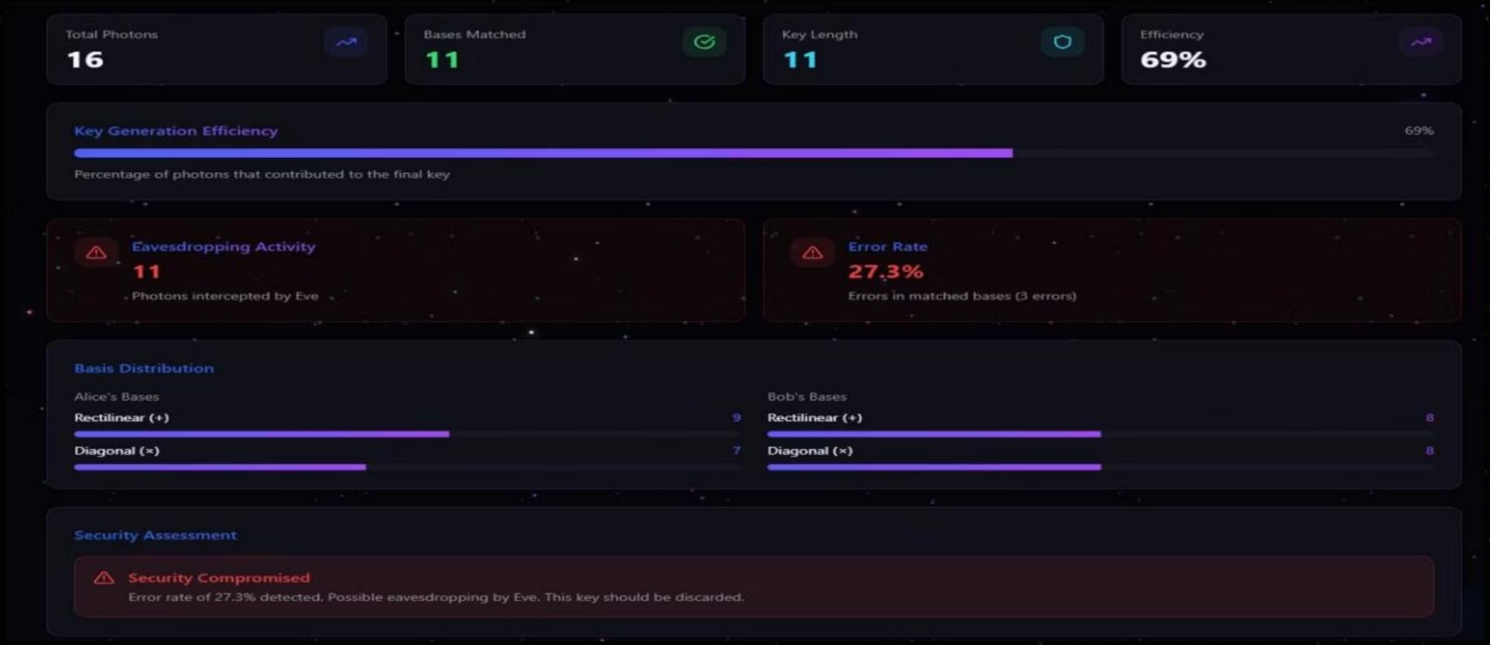
Key generation efficiency



Error Rate

3 errors in matched bases

Security Assessment: Security Compromised. Error rate of 27.3% detected—possible eavesdropping by Eve. This key should be discarded.



The software model demonstrates basis distribution between Alice and Bob using rectilinear (+) and diagonal (×) bases, with 11 photons intercepted by Eve affecting security.

The Future of Quantum Security

Real-World Deployments

Governments, banks, and tech companies in China, US, and Switzerland use QKD for secure voting, financial transactions, and long-distance communication.

Key Advantages

Immediate eavesdropping detection, stronger security than classical cryptography, protection against future quantum computer threats.

Challenges

High costs, distance restrictions, specialized hardware needs, environmental noise, and device imperfections limit large-scale deployment.

QKD will play a vital role in building future-proof cybersecurity infrastructures, offering scalable solutions for next-generation secure communication systems.

Thank You!!!

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